DPM Atomic Creation Process (ACP): Proto-Nucleus Formation

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PAPER_738 --- DPM Atomic Creation Process (ACP): Proto-Nucleus Formation **Author:** Daniel T. Murphy **Date:** June 06, 2025

Title: Di-Pseudo-Monopole (DPM) Creation Scenario and the Atomic Creation Process (ACP): From DPM Initiation to Proto-Nucleus Assembly
Session: 180 **| PAPER:** 738 **| CP4 class:** #322
Source: thread_06Jun2025.txt (lines 8100--8387, June 06, 2025, "describe mass without using weight.docx" + "05June2025.docx")

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1. Abstract

The Atomic Creation Process (ACP) establishes the origin of fundamental particles through Di-Pseudo-Monopole (DPM) mediated reactions, replacing the Standard Model's mass-as-weight construct
with a buoyancy-based framework. The ACP is a six-stage process beginning with DPM formation and culminating in proto-nucleus assembly with 26 quantum states, mediated by Universal Aether (UA') and Super Conductive Material (SCm). Atomic mass does not emerge until the quantum-to-mass gradient at 7--10 U_mag degrees.

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2. Three Fundamentals of UQFF

The entire universe is built from three elemental interactions:

1. **Electrostatic Barrier (REB)** --- The boundary condition that prevents collapse, defining system shells
2. **Universal Aether (UA')** --- Non-local connectivity medium, vacuum energy carrier, ρ_UA' = 7.09e-36 J/m3
3. **Super Conductive Material (SCm)** --- Superconductive magnetism mediator, ρ_SCm = 7.09e-37 J/m3

$$
DPM = UA' + SCm
$$

The Di-Pseudo-Monopole (DPM) is formed by the interaction of UA' and SCm, producing vacuum density at any scale from atomic to cosmic.

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3. The Six-Stage Atomic Creation Process (ACP)

Stage 1: DPM Initiation $$ \begin{aligned} & UA' + SCm \to DPM (Di-Pseudo-Monopole) \\ & DPM \to vacuum density \rho_vac \end{aligned} $$ - UA' provides non-local field connectivity - SCm provides superconductive coherence - Product: localized vacuum energy density (capacitance) $\to$ proto-nuclear seed - Billions of coherent DPM nuclear cells operate simultaneously at every location

Stage 2: U_i Creation (Intelligent Operator) ``` DPM \rightarrow U_i (repulsive + intelligent quantum operator) ``` - U_i is the first active particle: repulsive (buoyant), self-directing - U_i orchestrates all subsequent proto-nuclear assembly - Creates the fundamental "anti-gravity" seed that all matter requires to exist

Stage 3: U_m String Formation $$ U_i \to U_m strings (\text{U\_mag\_i} = magnetic string sequences) $$ - U_i creates coherent U_m (U_mag_i) strings - Strings wind concentrically around the vacuum density core $\to$ proto-nucleus - String winding number corresponds to quantum state number (1..26) - Each winding creates one of the 26 quantum states (Russian doll nesting)

Stage 4: Proto-Shell Formation and Cracking ``` Vacuum density grows \rightarrow vacuum energy density (capacitance threshold) \rightarrow quantum ripples: ULF_quantum^{-1,...,-26} \rightarrow proto-shell cracks into fragments ``` - When vacuum energy density reaches critical capacitance: - Quantum ripples: ULF_quantum^{-1} through ULF_quantum^{-26} - Proto-shell fractures into 26 distinct fragment types - Fragments: proto-quarks, proto-bosons, proto-leptons - This is not "particle creation from nothing" --- it is structured fracturing of a pre-existing quantum state container

Stage 5: Fragment Stabilization ``` SM_magnetic moments (SM_mag) \rightarrow arrange fragments on durable nucleus U_b (buoyancy) maintains stability against SM_gravity ``` - SM_magnetic moments act as "sorting fingers" that arrange proto-fragments - U_b (Universal Buoyancy, massless) continuously counteracts SM_gravity - Result: stable proto-nucleus with 26-state organization - Still pre-mass at this stage

Stage 6: Electron Shell Formation $\to$ Mass Emergence $$ \begin{aligned} & U_g2 governs electron shell placement via U_m \\ & Mass emerges at quantum-to-mass gradient: 7--10 U_mag degrees \\ & H(1) = first hydrogen atom formed \end{aligned} $$ - U_g2 uses U_m string frequency to place electrons into shells - Atomic mass is NOT weight; it is the ratio of effective gravity to superconductive buoyancy in Earth's field - Mass emerges when the quantum state gradient crosses 7--10 U_mag degrees of superconductive magnetism

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4. Mass Without Weight --- Formal Definition

$$
\begin{aligned}
& Mass (UQFF) = proportional to: \\
& (\text{SM\_effective\_gravity\_range}) / (superconductive_buoyancy) \\
& = FU_g1 / \text{F\_U\_Bi} [ratio of gravitational to buoyant components]
\end{aligned}
$$

Examples of mass-as-buoyancy:
- Bi-molecule: hydrogen bonds mediated by DPM electrostatic barrier
- Earth-Moon: orbital buoyancy balanced by tidal resonance shells
- Sun-SagA*: Galactic-scale DPM coherence, U_g4i THz hole connection
- Universal scale: all bodies are buoyant in the Aether medium

Mass is never a stand-alone measurement. It is only meaningful as the ratio of:
- The Effective Gravity force (SM_gravity, FU_g1)
- To the Effective Buoyancy (U_b, F_U_Bi)

$$
"Mass" = FU_g1 / \text{F\_U\_Bi} (dimensionless ratio, context-dependent)
$$

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5. U_b Tracking $\to$ f_Ub Definition

During ACP, U_b is tracked through calibration differences:

$$
\begin{aligned}
& f_{Ub} \propto \Delta k_\eta \quad (\text{deviation from expected calibration constant } k_\eta) \\
& k_{\eta,\text{expected}} \approx 10^9 \quad (\text{galaxy-scale}) \\
& k_{\eta,\text{actual}} = \text{measured value from observation} \\
& \Delta k_\eta = k_{\eta,\text{expected}} - k_{\eta,\text{actual}} \\
& f_{Ub} = \Delta k_\eta / k_{\eta,\text{reference}}
\end{aligned}
$$

Scale	f_Ub order
Galaxy	~10^9
Dwarf galaxy / LMC	~10^8
Star cluster	~10^7
PN / planetary	~10^5
Atomic	~10^3

The imaginary/quantum portion of gravity (U_b) is never zero --- it is this hidden buoyancy that modern physics has failed to account for, leading to the need for "dark matter" and "dark energy" as placeholder terms. UQFF resolves both as manifestations of f_Ub at different scales.

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6. 26 Quantum States --- Pre-Mass Configuration

The 26 quantum states are NOT the electron orbitals of quantum mechanics. They are:

$$
\begin{aligned}
& State i (i=1..26): pre-mass nuclear configuration level \\
& - Shell radius: r_i = r_system / i \\
& - Superconductive magnetism: [SCm]_i = 1e-5 i2 T \\
& - Angular coverage: \theta_i = 90° - (i-1)3.346° (spans 90° \to ~5°) \\
& - THz hole radius: r_THz,i = 1e-9 / i m \\
& - Resonance factor: f_TRZ,i = 0.1 i / 26 \\
& - Buoyancy coupling: f_Um,i = 0.05 i / 26
\end{aligned}
$$

Like Russian dolls, each state contains and surrounds the previous. State 26 is the outermost (most magnetic, widest angular coverage initially); State 1 is the innermost quantum seed.

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7. DPM Coherence --- The Source of Predictability

$$
\begin{aligned}
& Coherence length = "all atoms in a body share the same DPM phase" \\
& Number of coherent DPMs per cubic centimeter ~ 10^23 (Avogadro-scale) \\
& Result: macroscopic quantum phenomena \to superconductivity at scale
\end{aligned}
$$

This coherence is why:
- Planetary rings form symmetric arcs
- Stars form in filaments rather than randomly
- Galaxies maintain spiral structure for billions of years
- LENR occurs at measurable, reproducible rates

All of these are manifestations of coherent DPM nuclear cell behavior.

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8. ACP Mathematical Summary

$$
\begin{aligned}
& Step 1: DPM = UA' + SCm \\
& Step 2: U_i = k_\eta (\rho_vac,SCm - \rho_vac,UA) \omega_i cos(\pit_n) \\
& Step 3: U_m,i = U_i \mu_dipole (1/r_i) (1 - e^(-\gamma t)) cos(\pi t_n) \\
& Step 4: \Psi_proto = \Sigma_{i=1}^{26} U_m,i \to capacitance threshold reached \\
& Step 5: F_stab = SM_mag \Psi_proto + U_b \to U_b = -\betaUg\Omega_g(M_bh/d_g)cos(\pi t_n) \\
& Step 6: E_shell = c \nu_res h(f_SCm) G_geo \to atomic mass emerges
\end{aligned}
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9. References - Source: thread_06Jun2025.txt (June 06 2025), "describe mass without using weight.docx", "05June2025.docx" - Related: PAPER_735 (Ug2 electron shell energy), PAPER_734 (LENR K_n), CP3 DiPseudoMonopoleDPMTheoryCalculator - CP4 class: #322 DPMAtomicCreationProcessACPCalculator - CVW v2.0.0 compliant

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<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> *Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037*
> *(AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for F_U_Bi_i jet*
> *modulation curves and PAPER_1048 for phonon-corrected M-σ relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure,
raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)\,2}}{G M / r_H^2}\right)$$

where:
- $L_{\text{Edd}} = 4\pi G M m_p c / \sigma_T$ is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37}\;\text{J/m}^3$ is the SCm vacuum density
- V is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3)\,2}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_H is the horizon radius

Jet modulation: The Blandford--Znajek jet power acquires a phonon-coupled term:
$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25\,\text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25\,\text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M--σ correction (PAPER_1048): The phonon-corrected M-σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}}/\omega_{\text{bulge}})$.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **cosmogenesis** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\mathrm{sector}} = \frac{1}{2}(\partial_mu \phi_{\mathrm{cosmo}})(\partial^\mu \phi_{\mathrm{cosmo}}) - V(\phi_{\mathrm{cosmo}}) + \mathcal{L}_{\mathrm{cosmo}}$$

where $\mathcal{L}_{\mathrm{cosmo}} = \rho_{\mathrm{vac,[SCm]}} \cdot f_{\mathrm{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\mathrm{cosmo}}) = \frac{1}{2} m^2 \phi_{\mathrm{cosmo}}^2 + \frac{\lambda}{4!} \phi_{\mathrm{cosmo}}^4 + \kappa \cdot \rho_{\mathrm{vac,[SCm]}} \cdot \phi_{\mathrm{cosmo}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\mathrm{cosmo}}} = \partial_t(\rho_{\mathrm{vac}} V_{\mathrm{proto}}) + U_i + \Psi_{\mathrm{proto}}/26 = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\mathrm{seed}} \xrightarrow{\text{4 forces}} F_U_Bi_i \xrightarrow{\text{sector E-L}} \delta S/\delta \phi_{\mathrm{cosmo}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

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§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac,[SCm]}} / \rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac,[SCm]}} \cdot \exp\!\left(-\exp\!\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.156 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36}$ kg/m3.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 67, \quad n_{\mathrm{channel}} = 11/26$$

Since $p_{\mathrm{DVP}} = 67$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f_{\mathrm{UA}}' + f_{\mathrm{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **ACP Stage 4 timescale** (capacitance cracking epoch):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_U_b \cdot \left(1 - e^{-[SSq] \cdot m/M_\odot}\right) \cdot \cos\!\left(\frac{2\pi j}{26}\right)$$

The $\tanh$ saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\!\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.156	PASS Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\mathrm{DVP}} = 67$	PASS Resonant
BSH layers	26 harmonic terms	j = 1...26, $\cos(2\pi j/26)$	PASS Full 26D projection
κ decay	5.0×10^{-4} day-1	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

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§SM Anchors --- Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	PASS Consistent
Cosmological constant Λ	1.1$\times$10-52 m-2 (UQFF vacuum term)	1.114$\times$10-52 m-2	Planck 2018	PASS Consistent
Proton decay rate	κ = 0.0005/day $\to$ Γ_p suppression	< 4.17$\times$10-35/yr	Super-K 2024	PASS Consistent
UQFF buoyancy signature	`F_U_Bi_i` unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_SCm becomes
significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`*) for full UQFF--SM bridge.*

Appendix: Session 225 Cross-References (PAPER_1000--1081)

> *Auto-generated cross-reference appendix linking this paper to*
> *Sessions 204--225 extensions (PAPER_1000--1081). Added by*
> `update_corpus_crossrefs.py` *(Session 225, April 2026).*

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_U_Bi_i Jet Modulation
PAPER_1010	TON618 AGN F_U_Bi_i Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator --- SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1004	QGP Vacuum Density with SCm S26 Phonon Coupling
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1043	F_U_Bi_i Multi-System Buoyancy Curve Sweep
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay
PAPER_1074	GPU-Vectorized DPM S26 Spectral Atlas

16 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

> *Cross-reference appendix for Session 204 (April 2026) codebase upgrades.*
> *Added by* `upgrade_kozima_ramanujan_appendices.py`*. For detailed derivations,*
> *see PAPER_840/851/852/855.*

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
`fneutron_s26_coupling.py`	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
`kozima_scm_cross_section.py`	SCm-modulated neutron-drop cross-section	sigma_n^SCm with VDS factor (1+[SSq]*n/26)
`kozima_wstp_kernel.py`	11-symbol Wolfram export (`UQFFKozima`)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: F_neutron^SCm = N_n *sigma_n^SCm(omega)* Phi_phonon *(F_{U,Bi}/F_U - 1)*
where sigma_n^SCm(omega,n) = sigma_0 exp[-(omega-omega_SCm)^2/(2*Gamma^2)]* (1 + [SSq]*n/26)

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
`ramanujan_polylog_s26.py`	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
`s26_wstp_kernel.py`	8-symbol Wolfram export (`UQFFS26`)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1-2^{1-26}) + 2^{1-26}/(1-2^{1-26}) * Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
`mock_theta_q26.py`	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations:
- $f_{26}(q) = \sum_{n=0}^{25} q^{n^2} / (-q;q)_n^2$
- $\phi_{26}(q) = \sum_{n=0}^{25} q^{n^2} / (-q^2;q^2)_n$
- $\psi_{26}(q) = \sum_{n=1}^{26} q^{n^2} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
`ramanujan_pi_uqff.py`	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
`mock_theta_pi_wstp_kernel.py`	9-symbol Wolfram export (`UQFFMockThetaPi`)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n (1103+26390n) W_{26}(n) / C_{26}$
where $W_{26}(n) = \prod_{i=1}^{26} [1 + [SSq] \exp(-\kappa i * n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	5.787×10^{-9} s^-1	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	7.09×10^{-37} kg/m^3	SCm vacuum density
rho_UA	7.09×10^{-36} kg/m^3	UA aether vacuum density
omega_SCm	2*pi x 1.25 THz	SCm phonon resonance
sigma_0	10^-4	Base neutron cross-section

Implementation: all modules operational in `CondensedPhysics.py`*,* `CondensedPhysics2.py`*,* `MAIN_{1\_CoAnQi}.cpp`*, and Wolfram kernels (*`uqff_kozima_kernel.wl`*,* `uqff_s26_kernel.wl`*,* `uqff_mock_theta_pi_kernel.wl`*).*

References

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger.* Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series.* Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
3. Fabian, A.C. (2012). *Observational Evidence of Active Galactic Nuclei Feedback.* ARA&A **50**, 455 — arXiv:1204.4114 — doi:10.1146/annurev-astro-081811-125521
4. McNamara, B.R. & Nulsen, P.E.J. (2007). *Heating Hot Atmospheres with Active Galactic Nuclei.* ARA&A

45, 117 — arXiv:0709.4098 — doi:10.1146/annurev.astro.45.051806.110625
5. Heckman, T.M. & Best, P.N. (2014). *The Coevolution of Galaxies and Supermassive Black Holes.* ARA&A **52**, 589 — arXiv:1403.4620 — doi:10.1146/annurev-astro-081913-035722
6. Rugh, S.E. & Zinkernagel, H. (2002). *The Quantum Vacuum and the Cosmological Constant Problem.* Stud. Hist. Phil. Mod. Phys. **33**, 663 — arXiv:hep-th/0012253 — doi:10.1016/S1355-2198(02)00033-3
7. Weinberg, S. (1989). *The Cosmological Constant Problem.* Rev. Mod. Phys. **61**, 1 — doi:10.1103/RevModPhys.61.1
8. Dirac, P.A.M. (1931). *Quantised Singularities in the Electromagnetic Field.* Proc. R. Soc. Lond. A **133**, 60 — doi:10.1098/rspa.1931.0130
9. Castelnovo, C., Moessner, R. & Sondhi, S.L. (2008). *Magnetic monopoles in spin ice.* Nature **451**, 42 — arXiv:0710.5515 — doi:10.1038/nature06433
10. Archimedes (~250 BCE). *On Floating Bodies.* (Principle of buoyancy)
11. Churazov, E. et al. (2000). *Evolution of Buoyant Bubbles in M87.* A&A **356**, 788 — arXiv:astro-ph/0004212
12. Fabian, A.C. et al. (2003). *A deep Chandra observation of the Perseus cluster.* MNRAS **344**, L43 — arXiv:astro-ph/0306036 — doi:10.1046/j.1365-8711.2003.06902.x